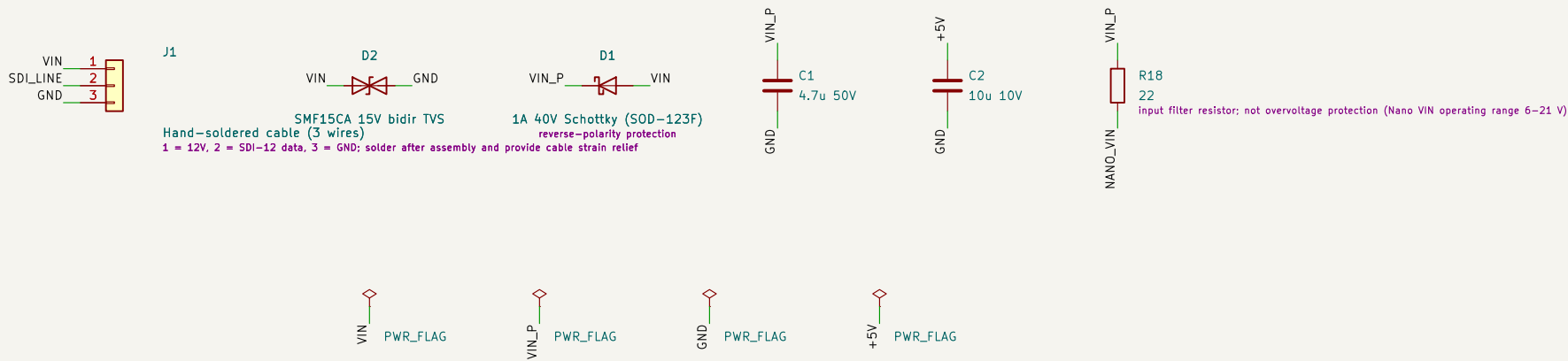


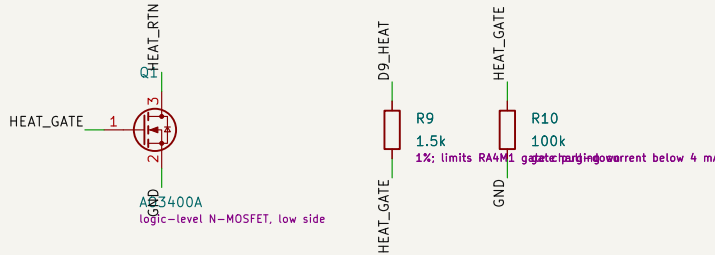
POWER IN — 12 V battery / logger supply, normally 11.5–14.4 V, through three soldered wires. D1 provides reverse–polarity protection. The Nano VIN branch uses R18; its onboard regulator supplies the hat's 5 V. The TVS and R18 do not guarantee a 21 V surge limit.

SDI–12 DATA — R3 limits transmit and fault current; R4 sets the idle load. C4 slows transitions and D4 provides ESD protection. Direct 5 V GPIO receive thresholds are not guaranteed for every compliant SDI–12 sender; verify with the intended logger and cable.

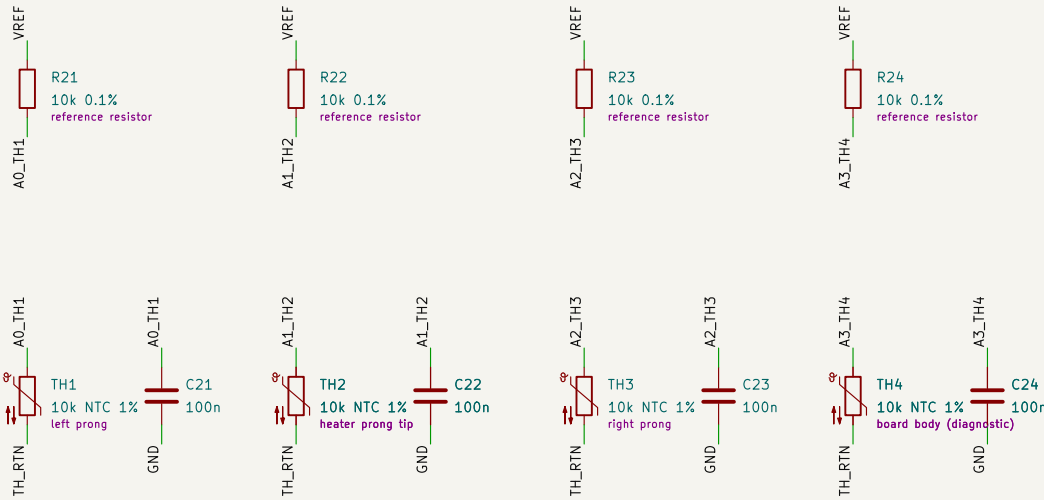
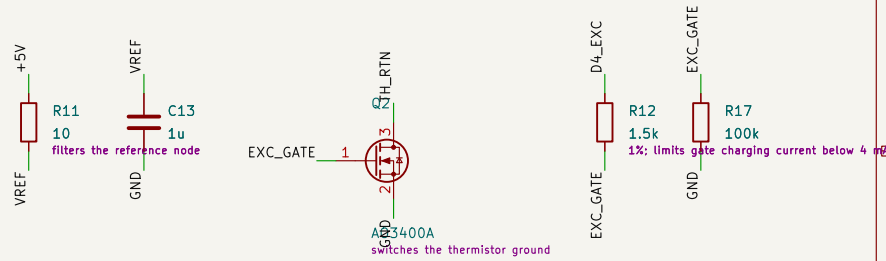
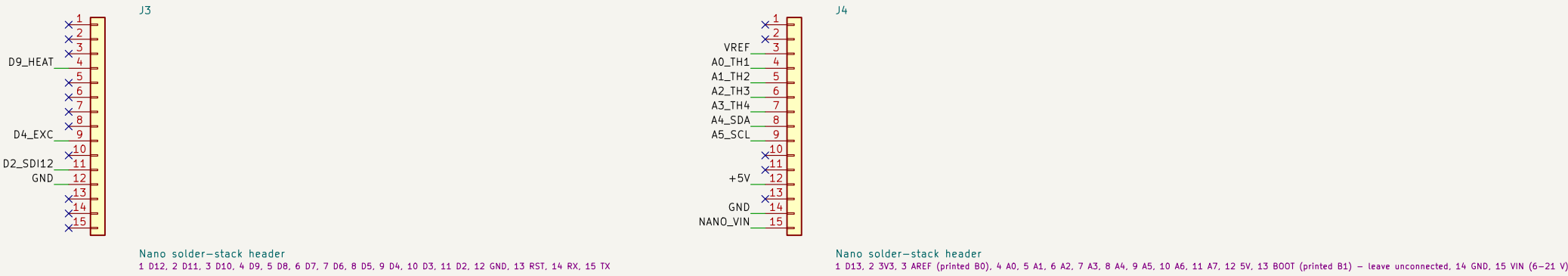
HEATER DRIVE + POWER MONITOR — heater runs from the battery, never from USB alone. Firmware must bound pulse duration to 15 s, monitor V and I, and switch OFF on timeout or a fault. Refuse heating above 15 V input. R10 holds the gate low when D9 is high impedance; it is not a watchdog.



ARDUINO NANO R4 ABX00142 — two Samtec HTSW–115–07–T–S 1x15 strips soldered through BOTH boards; components face inward. Seat header insulators on hat front and use a fixture to hold a 4.00 mm board gap during soldering. Trim solid pin tails to 0.5 mm maximum outside each board. USB–C faces the cable end. Loose pins are not reliable contacts. Program using Arduino Nano R4 in the IDE before potting.



THERMISTORS — dividers and Nano AREF share filtered VREF. Q2 switches their ground return to reduce self–heating between readings. Firmware: analogReference(AR_EXTERNAL), analogReadResolution(14); D4 HIGH, wait 10 ms, sample, D4 LOW. Verify settling, noise and calibration on assembled hardware. NTC = Murata NCP15XH103F03RC (10k, 1%); Rref = 10k 0.1% 25 ppm.



HEATER RESISTOR CHAIN (top side of the centre prong only) — 17 x 3.3 ohm 1% 0603 in series, 2.6 mm pitch. EXACT Vishay CRCW06033R30FKEAHP, 0.33 W at 70 C. The return runs on the bottom copper of the prong. Use Panasonic ERJ2GEJ5R1X, 0.1 W; resistor tolerance and local temperature affect the margin. First resistor starts outside the epoxy body. Firmware must detect zero heater current as a fault.

